SME Technology Development (R&D) Support Project Research and Development Plan

Project Overview

Project Title	Development of a RAG-based Business-Specialized Real-time Multilingual Translation and Interpretation AI Solution - A Global Sales, Consultation, Interpretation, and Documentation Automation Integrated Platform for Export Revenue Innovation for SMEs
Development Goals	Implementation of Business-Specialized AI Model: Develop a RAG-based Multilingual Translation and Interpretation Model trained on unique enterprise knowledge (product names, technical specifications, prices, etc.) and an End-to-End (E2E) real-time streaming voice interpretation model (for more than 8 languages) with a delay of less than 2 seconds. Full-cycle automation for export operations: Develop 5 types of export document automatic generation agents such as quotations and product pages. Unique Regulatory Certification Formatting: Build a SaaS platform integrating complex regulatory certification document formatting modules such as eCTD, KC, CE, FCC to eliminate global entry barriers for SMEs and achieve sales innovation
Development Method	[Core Differentiating Data] Utilize an exclusive corpus of 1.1 million sentence pairs (600,000 business + 500,000 regulatory certifications) that cannot be replicated in a short period for domain-specialized fine-tuning to establish a formidable entry barrier. [AI Model Enhancement] Combine RAG technology injecting enterprise-specific knowledge into NLLB/EXAONE-based hybrid language models to reduce hallucinations by 70%. Secure simultaneous interpretation performance within 2-second delay using InfiniSST and Whisper-based technology. [SaaS Platform Commercialization] Develop a cloud SaaS web, mobile, API integrated environment leveraging the validated AI platform (Oz-i) infrastructure of the leading agency to enable user companies to link and use their data within three days without IT personnel.

Research Team Composition	[Lead] PCN Co., Ltd. (Platform/Commercialization Management): An AI company with 25 years of experience (270 billion KRW sales) having GS-certified platforms, possessing the Oz-i AI platform, and extensive know-how in various R&D projects, ensuring SaaS platform implementation, API integration, and cloud infrastructure commercialization. [Co-Leader①] Stunning Box Co., Ltd. (AI Model/Data Focus): An AI translation data company with 15 years of accumulated specialized datasets, responsible for AI translation fine-tuning and provision of proprietary corpus as a core asset, and multi-agent regulatory certification formatting technology. [Co-Leader②] 3 Demand Enterprises (Field Demonstration Verification): Companies awarded for excellence in exports such as Sunlight Ilukom Co., Ltd. (Automotive Parts), Genuine Grip Co., Ltd. (Special Shoes), Grey Whale (K-Beauty), provide empirical data for 3 industries and quantitatively verify the adoption effect through A/B testing.
Expected Effect	[Economic and Industrial Impact] Improve the export bidding success rate of user companies by over 20% and markedly reduce the regulatory certification document processing period from the current 4-8 weeks to 1-2 weeks. Generate a revenue of 22 billion KRW in the 5th year of commercialization (cumulative 68 billion KRW over 5 years), create 51 new jobs, and plan to distribute to over 500 SMEs through AI vouchers and KOTRA linkage. Achieve a cumulative export substitution effect of 44 billion KRW over 5 years. [Securing World-Class Technology] Attain over 65 BLEU in translation accuracy, less than 2-second delay in voice interpretation, and over 70% document structure preservation rate (TEDS) to capture the B2B domain-specialized translation and interpretation market unaddressed by global big tech (Google, DeepL).

Self-Diagnosis

Project Title	Global Sales, Consultation, Interpretation, and Documentation Automation Integrated Platform for Export Revenue Innovation for SMEs	
Check Items	**해당 있음**	**해당 없음**
1. Does the current applicant (research and development institution, etc., hereinafter referred to as the 'applicant') not meet the application qualifications set in the project announcement, etc.?		✓
2. Are there any related documents required by the project announcement, etc., that the current applicant has not submitted?		✓
3. Is the current applicant under a restriction on participation in national R&D?		✓
4. Does the applicant have any history of sanctions, such as restrictions on participating in government R&D, due to misemployment of labor costs or false transactions?		✓
5. Is the applicant currently a bankrupt, temporarily closed, or shut-down company?		✓
6. Has the applicant ever been subject to disposition for tax arrears on national, local, or customs taxes by tax authorities, or has information such as arrears of fines, industrial accident, or employment insurance arrears been registered with a comprehensive credit information institution?		✓
7. Is the applicant currently registered as a defaulter in the debtor registry under the Civil Execution Act or as a defaulter with a comprehensive credit information institution?		✓
8. Has an application for the commencement of bankruptcy, rehabilitation procedure, or personal rehabilitation procedure been filed with the court for the applicant?		✓
9. Is the applicant recently in a state of debt ratio over 1,000% or in a completely capital-impaired status based on the most recent year's financial statements?		✓
10. As a result of self-assessment, does the project applied for fall under the classification criteria of security projects as per Article 45, Paragraph 1 of the Enforcement Decree of the Innovation Act?		✓

※ In addition to the items specified in the above self-diagnosis table, you must meet the eligibility requirements stipulated in the business announcement and related regulations.

Based on the review of these matters, you may be excluded from support or have your selection canceled.

The criteria for "current" in the above confirmation items refer to the qualification requirements of the business announcement and related regulations on the deadline for receipt.

As of the date determined for verification purposes

Prepared by each leading and cooperating institution

The applicant has verified the above details, and if they are found to be inaccurate I have no objection to the possibility of exclusion from support or cancellation of selection.

Year 2026 July Day

Lead Institution Applicant: Sanghyun Um

Lead Institution Corporation Name: PCN Co., Ltd.

1. Technical Development Goals



1) Background and Necessity of Business Promotion

(1) 환경 분석

Export SMEs face complex barriers such as translations not reflecting company-specific work contexts in overseas sales processes, unsuitable interpretation structures for real-time consultations, and high cost/long-duration regulatory documentation compliance. Therefore, an AI platform specialized in domain-specific tasks that connects the entire export cycle with company data and terminology beyond mere translation features is necessary.

Key Barriers	On-site Issues	Need for Technology Development
① Work Context and Terminology	General-purpose AI does not adequately reflect company-specific product names, technical specifications, pricing, and contract terms, risking critical information mistranslation during negotiation and contract processes.	Integration of company-specific product DBs, price lists, and glossaries with RAG to reflect translation, interpretation, and document generation in real-time.
② Real-time Interpretation Performance	The existing 3-step structure of STT→MT→TTS accumulates delays of 2~5 seconds or more and has limitations in handling background noise, speaker separation, and specialized terminology.	Implement real-time interpretation within 2 seconds using E2E streaming architecture with speaker separation and specialized terminology injection.
③ Regulatory Certification Documents	CE, eCTD, FCC, HACCP, etc., have complex country-specific formats and document structures, with preparation time and outsourcing costs being a significant burden for SMEs.	Automate translation, structuring, and formatting of regulatory documents and support the creation of submission documents based on expert review (HITL).

(2) Market Growth and Urgency of Development

◦ The global business AI translation market is expected to grow from approximately $1.8 billion in 2025 to over $7 billion by 2030, with demand for business domain specialization rapidly expanding.